Towards a real-time vehicular emissions inventory

IAG SCIENCE DAY

Dr. Sergio Ibarra-Espinosa
Post-doc, DCA/IAG/USP
sergio.ibarra@usp.br
http://github.com/ibarraespinosa/

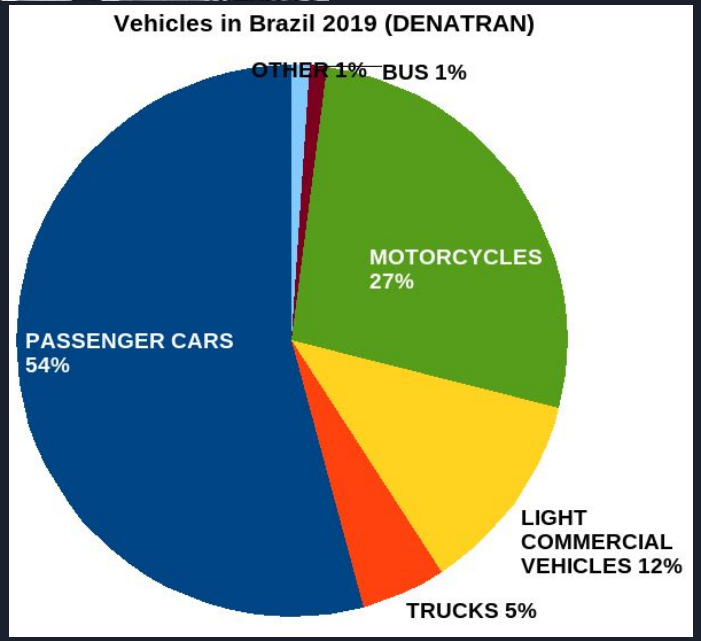
Introduction

- Brazilian fleet has experienced a dramatic change in the last decades Currently, there are:

102,666,444

vehicles registered in Brazil (DENATRAN 2019)

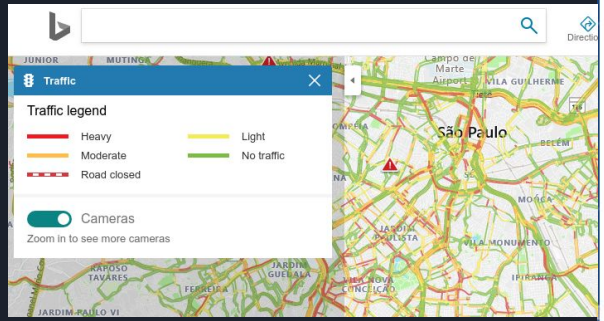
- Emissions of **NOx, CO, PM, VOC** and more pollutants affects air quality with deleterious effects on human and ecosystem health
- **Emerging technologies and bigdata can be used to develop real-time vehicular emissions inventories**

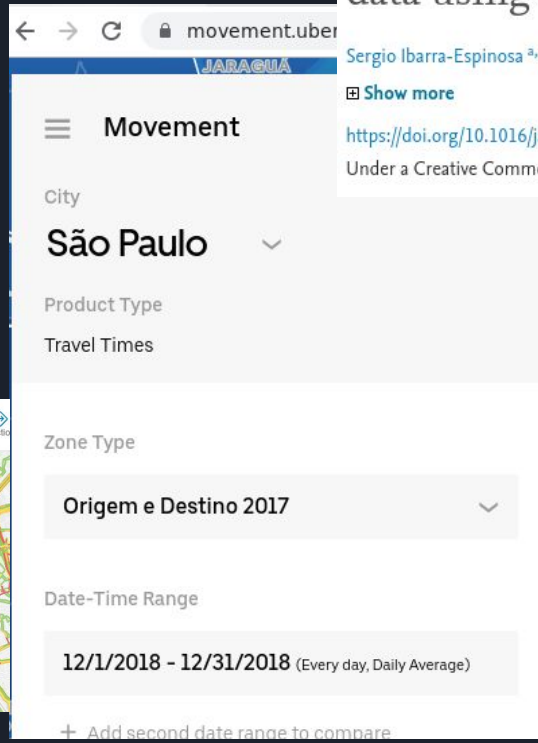


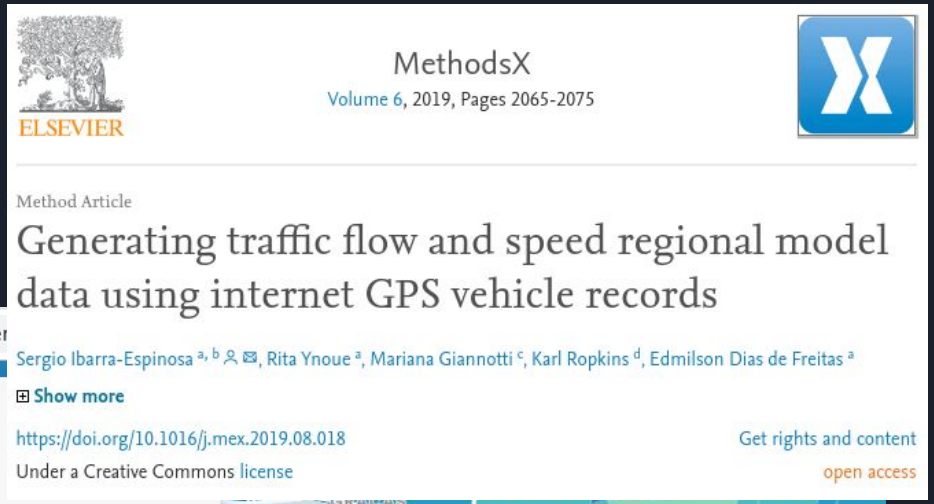
Picture Brazil 1956: Movimento de carros na Avenida Rubem Berta (1956) https://br.pinterest.com/pin/74731613188fu7974437/
Electric car sold in Brazil 2019: https://revistaautoesporte.globo.com/Noticias/noticia/2018/06/unico-carro-eletrico-venda-no-brasil-bmw-i3-reestilizado-tem-preco-de-r-199950.html

Real Time traffic data

- GPS, radar and Smartphones
- Uber, Waze, Didi, Baidu Maps provide average speed



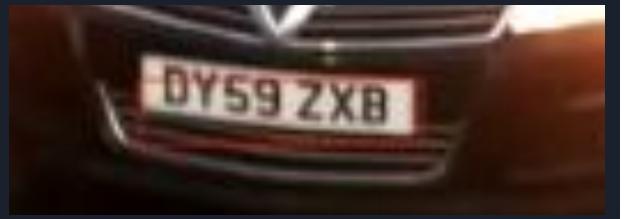


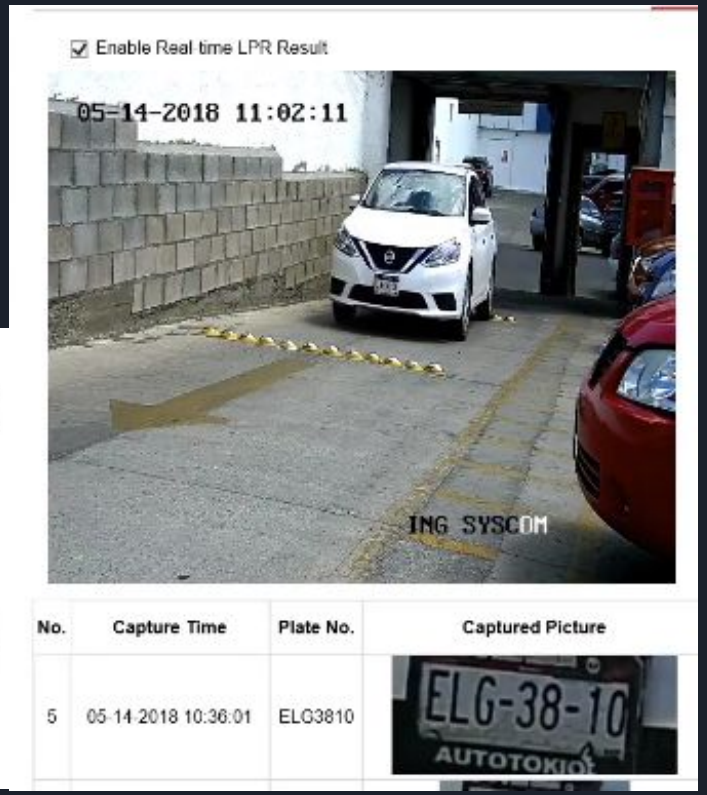


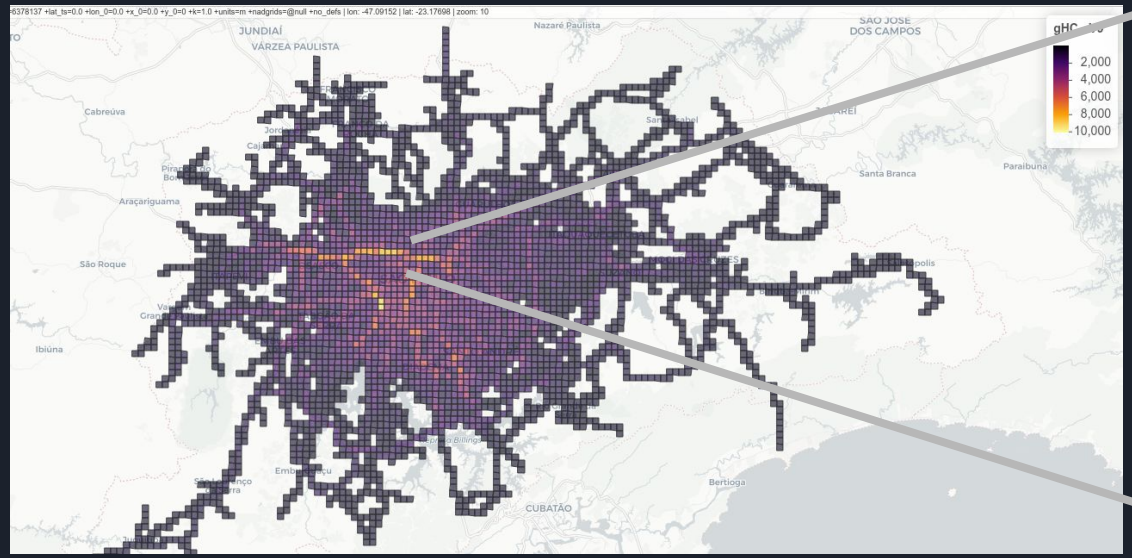
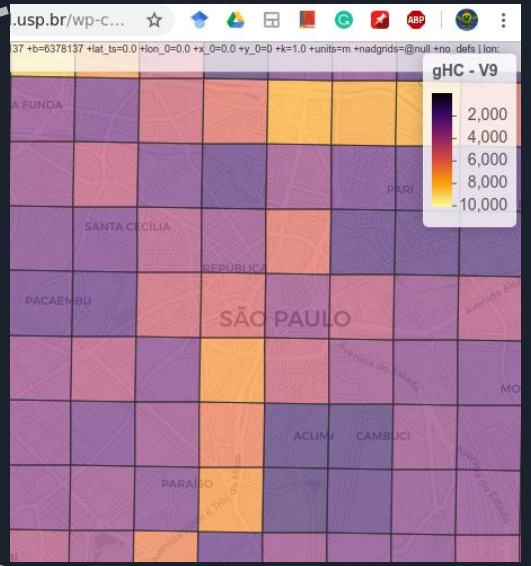
Real Time traffic data - OCR + OpenCV

https://www.youtube.com/watch?v=XszBqvQ2XCg 1 min

- Optical character recognition (OCR) Tesseract OCR open-source C++
- OpenCV a C++ AI library





https://www.youtube.com/watch?v=nQJTs6j4jVk
https://www.youtube.com/watch?v=rik8WOPNdSY

Real-time environmental evaluation for smat cities